

Sanjith Ganapathi

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EDUCATION

International Institute of Information Technology, Hyderabad

Hyderabad, India

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

CGPA: 8.68

EXPERIENCE

Undergraduate Researcher, CVEST Lab, IIITH | *Advisor Prof. Priyesh Shukla* April 2026 – Present

- Co-authored a paper accepted at EdgeSys 2026 on hardware-aware benchmarking of VLM inference on Qualcomm SM8750, characterizing CPU/NPU execution phases, latency, energy consumption, and thermal behaviour.
- Currently working on KV Cache Optimization, In-Context Learning.

Care Plan Driven Intervention Management System (Optigene.ai) | *TypeScript, JWT, Bun, Elysia.js, PostgreSQL*

- Built a multi-tenant healthcare intervention platform facilitating patient organization, care-plan creation, approval workflows, JWT based auth, role-based access, PostgreSQL-backed intervention pipelines
- Designed transactional intervention claiming system using PostgreSQL's row level locks for concurrent operators
- Deployed the application on a production VM and validated functionality through unit, integration, regression, and user acceptance testing.

KEY PROJECTS

AI Powered Recruitment System | *Python, Next.js, FastAPI, LangChain, ChromaDB, OpenAI/Ollama* | [GitHub](#)

- Built a full-stack recruitment assistant for resume ingestion, top-k candidate ranking, policy-grounded Q&A, and onboarding plan generation using FastAPI and Next.js.
- Implemented hybrid candidate retrieval with ChromaDB semantic search, BM25 keyword scoring, nice to have filters, and optional LLM-based query enhancement, reranking via LangChain and OpenAI/Ollama.

Distributed File System | *C, TCP Sockets, POSIX Threads, File I/O* | [GitHub](#)

- Designed a scalable DFS with a central Name Server and multiple Storage Servers, supporting concurrent access and granular sentence-level locking for over multiple users.
- Enforced persistent metadata management and an Access Control List to enforce user permissions for reading, writing, and executing file-based shell commands.
- Developed Storage Server-level Undo mechanism to revert the most recent changes across the distributed network.

xv6 Kernel: Demand Paging, Swapping & CFS Scheduler | *C, xv6, RISC-V, Operating Systems* | [GitHub](#)

- Modified the xv6 kernel to implement Lazy Allocation, FIFO page replacement, per-process swap system along with addition of FCFS, CFS schedulers with a fairness increase from the default round-robin scheduler.

C-Shell: Unix Shell with Job Control | *C, POSIX, Linux Systems Programming* | [GitHub](#)

- Built a POSIX-compliant Unix shell supporting command parsing, execution of external programs, pipelines, I/O redirection, sequential/background execution, and custom built-in commands.
- Implemented Unix job control with foreground/background process management, process groups, persistent command history, and signal handling (SIGINT, SIGTSTP, SIGCONT) using POSIX system calls

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, Bash, JavaScript, HTML/CSS, Assembly(x86-64)

Libraries/Frameworks: libpcap, NumPy, Matplotlib, FastAPI, Streamlit, PyDantic, Elysia.js

Backend and Systems: PostgreSQL, REST APIs, TCP Sockets, POSIX Threads, Linux Systems Programming

Developer Tools: Git, Docker, Linux, NeoVim, VS Code, WireShark

Core CS: Data Structures and Algorithms, Object-Oriented Programming, Design and Analysis of Software Systems, Design and Analysis of Algorithms, Operating Systems and Networks, Data and Applications

ACHIEVEMENTS

JEE Mains 2024: AIR 1084 (99.94 Percentile) **JEE Advanced 2024:** AIR 2738 **KCET:** Rank 87 (Karnataka)

Boards: 10th(96.6%) 12th(95.6%)

Sports: Bronze in National Inline Hockey Championship (Karnataka team captain)